

Analytical Assessment

Let the domain for x be all students in university

Let the domain for y be all programming languages

Let $L(x, y)$ be the predicate: "Student x knows programming language y ."

Translate the following two formal logic statements into clear, precise English sentences that clearly show the different in their meanings:

1. $\forall x \exists y L(x, y)$

2. $\exists y \forall x L(x, y)$

Solution:

1- $\forall x \exists y L(x, y)$

Means: For every student in the university
Knows at least one programming language.

2- $\exists y \forall x L(x, y)$

Means: There is a programming language that all
students in the university know.